

Assignment 2: (Fall 2013)

PHYSICS (PHY101)

TOTAL MARKS: 35

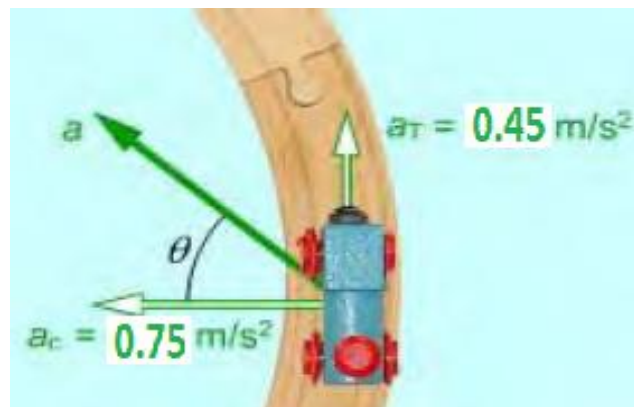
Due Date: 09/12/2013

DON'T MISS THESE *Important instructions*:

- ✚ To solve this assignment, you should have good command over first 15 lectures.
- ✚ Upload assignments (Microsoft word) properly through LMS, (No Assignment will be accepted through email).
- ✚ Write your ID on the top of your solution file.
- ✚ All students are directed to use the font and style of text as is used in this document.
- ✚ Don't use colorful back grounds in your solution files.
- ✚ Use Math Type or Equation Editor etc for mathematical symbols.
- ✚ Write to the point and avoid from unnecessary explanation. Don't copy/paste from internet or any other source.
- ✚ This is not a group assignment, it is an individual assignment so be careful and avoid copying others' work. If some assignment is found to be copy of some other, both will be awarded zero marks. It also suggests you to keep your assignment safe from others. No excuse will be accepted by anyone if found to be copying or letting others copy.

Don't wait for the last date to submit your assignment.

Question # 1



Find out the magnitude and direction of the overall acceleration in the fig shown? **Marks = 6**

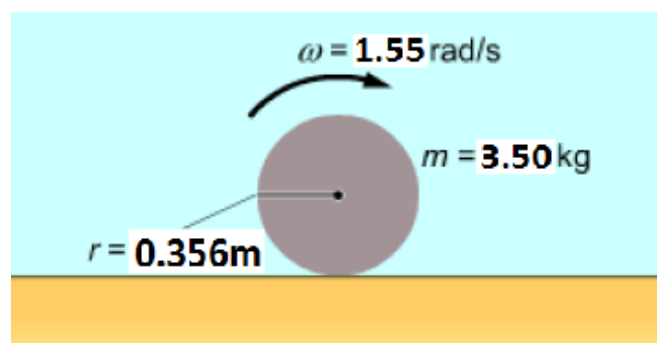
Question # 2



In its natural state, an average of $5.5 \times 10^6 \text{ Kg}$ of water flowed per second over Niagara Falls, falling 55m. If all the work done by gravity could be converted into electric power as the water fall to the bottom, how much power would the falls generate?

Marks = 8

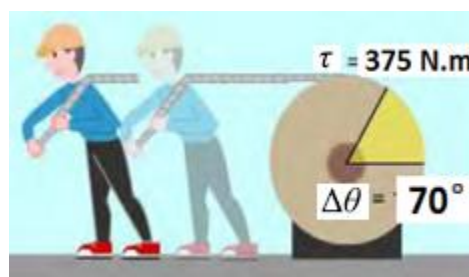
Question # 3



The solid uniform disk rolls without slipping. What is its kinetic energy as shown figure?

Marks =8

Question # 4



How much work is done when the wheel rotates shown in the figure?

Marks = 6

Question # 5



The tunable shown above makes a record rotate at 33.3 revolutions per minute. If the linear speed of an ant located at radius r is v , what is the linear speed of a point on the outer edge of the record at radius $3r$? **Marks = 3+4**

- also v
- $1/3v$
- $3v$
- $9v$
- $6v$

Write the reason of your above selected choice as well.

<<<<<<<<<<.....Good Luck.....>>>>>>>>